INCH-POUND

MIL-C-83522/15
19 December 1989

MILITARY SPECIFICATION SHEET

CONNECTORS, FIBER OPTIC, PLUG-RECEPTACLE-ADAPTER STYLE, FIXED SINGLE TERMINUS ENVIRONMENTAL FSMA STYLE, CRIMP AND CLEAVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-83522.

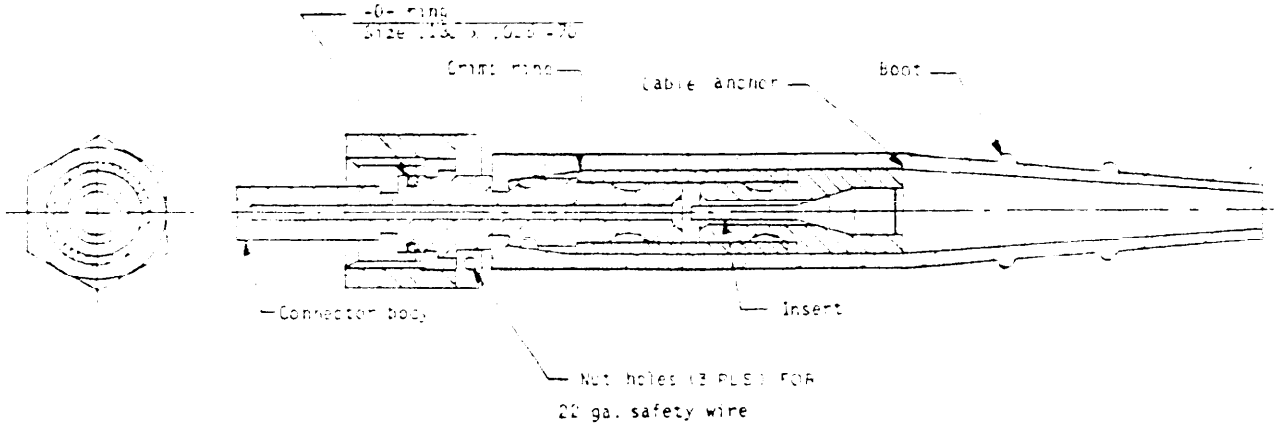
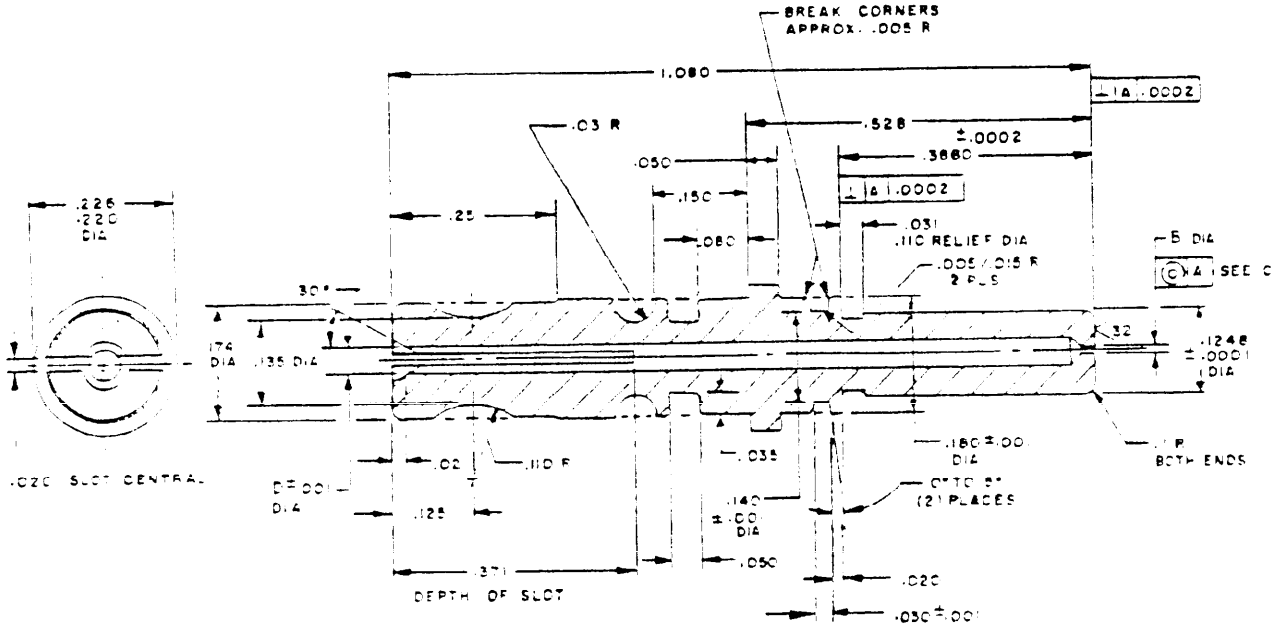


FIGURE 1. Assembled connector.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Dash number	Fiber core size	B diameter	C T.I.R.	D diameter
-1	200 µm	.230 +.004, -.001	.0002 inch	.042 inch
-2	400 µm	.430 +.010, -.000	.0004 inch	.042 inch

FIGURE 2. Connector body.

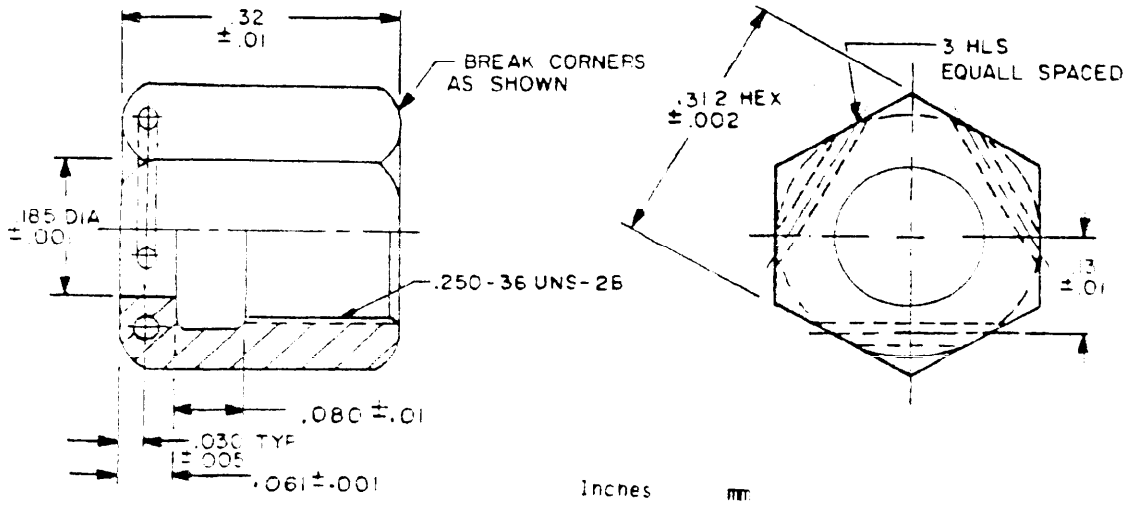
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Inches	mm	Inches	mm
.0001	0.003	.110	2.79
.0002	0.005	.1248	3.170
.0004	0.010	.125	3.18
.001	0.03	.135	3.43
.005	0.13	.140	3.56
.015	0.38	.150	3.81
.0156	0.396	.174	4.42
.020	0.51	.180	4.57
.030	0.76	.220	5.59
.0313	0.795	.226	5.74
.035	0.89	.250	6.35
.042	1.07	.371	9.42
.050	1.27	.3860	9.804
.080	2.03	.528	13.41
.086	2.18	1.080	27.43

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Break all sharp edges.
4. 125/finish or as noted.

FIGURE 2. Connector body - Continued.

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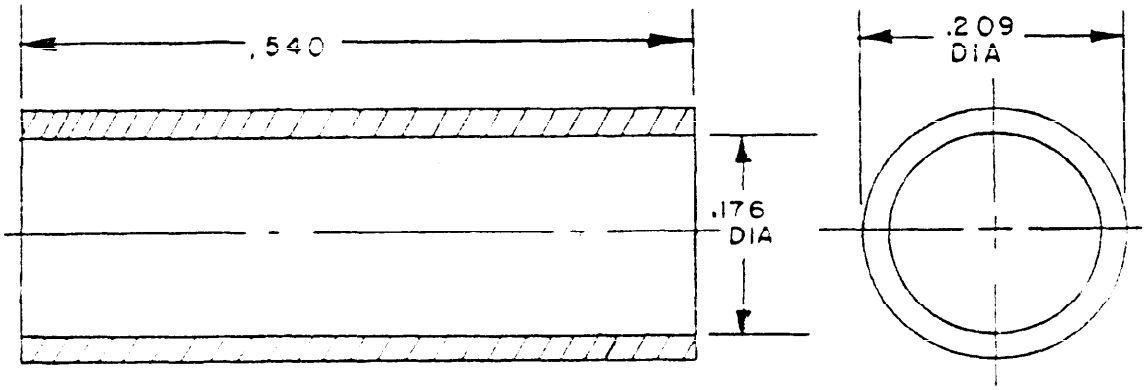


Inches	mm
.001	0.03
.002	0.05
.005	0.13
.01	0.3
.030	0.76
.061	1.55
.08	2.03
.13	3.3
.185	4.70
.250	6.35
.312	7.92
.32	8.1

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Break all sharp edges.
4. 125/finish or as noted.

FIGURE 3. Coupling nut.

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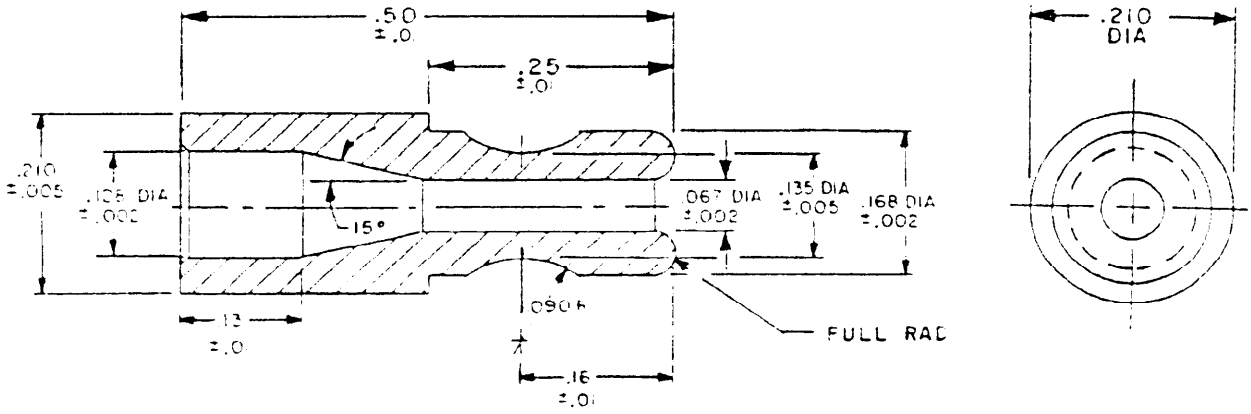


Inches	mm
.176	4.47
.209	5.31
.540	13.72

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given general information only.
3. Break all sharp edges.

FIGURE 4. Crimp ring.

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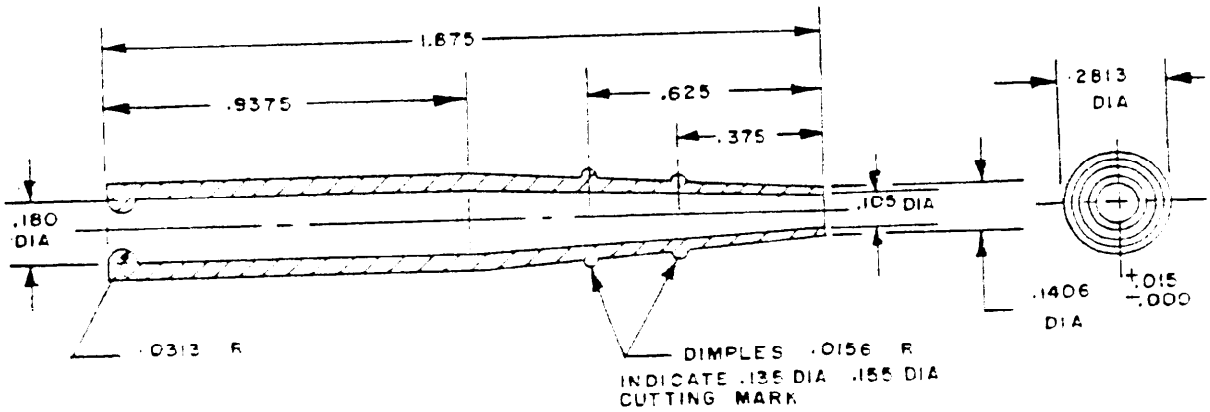


Inches	mm
.067	1.70
.090	2.29
.125	3.18
.135	3.43
.1563	3.970
.168	4.27
.250	6.35
.500	12.70

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given general information only.
3. Break all sharp edges.
4. All dimensions are after plating.
5. 125/finish or as noted

FIGURE 5. Cable anchor.

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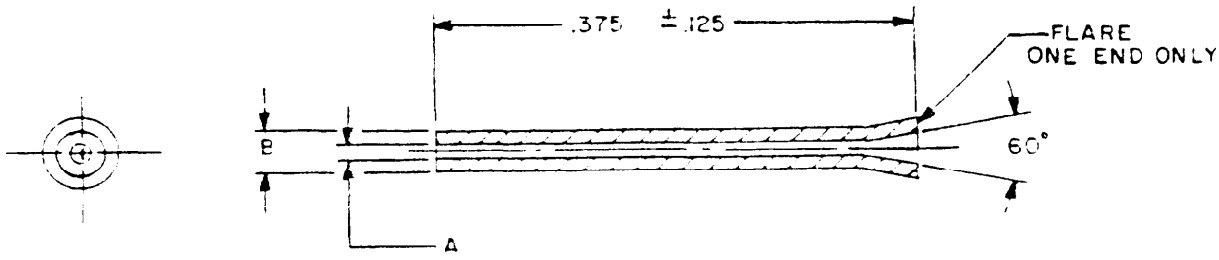


Inches	mm
.015	0.38
.0156	0.396
.0313	0.795
.105	2.67
.135	3.43
.1406	3.571
.155	3.94
.180	4.57
.2813	7.145
.375	9.53
.625	15.88
.9375	23.813
1.875	47.63

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 6. Boot.

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Inches	mm
.018	0.46
.025	0.64
.041	1.04
.125	3.18
.375	9.52

Dash number	A diameter ±.003	B diameter ±.001	Fiber core
-1	.018	.041	200 µm
-2	.025	.041	400 µm

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 7. Insert.

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REQUIREMENTS:

Metals: The plug housing shall be of corrosion resistant steel in accordance with QQ-S-763, class 303 or nickel plated brass in accordance with QQ-B-626. Other metal components shall be of corrosion resistant steel in accordance with QQ-S-763, class 303.

Dimensions and configuration: See figures 2 through 7.

Fiber optic cable requirements:

Cable diameter: In accordance with DOD-C-85045.
Cable configuration: In accordance with DOD-C-85045.
Fiber diameter: 200/230µm, 400/430µm.
Fiber numerical aperture: In accordance with DOD-F-49291.
Fiber type: In accordance with DOD-F-49291.

Cleaning procedures: Dampen lens tissue (paper wipes or cotton swab) with a small amount of isopropyl alcohol or water. Gently wipe the face of the connector, removing any debris, particularly around the optical fiber, using clean tissue (paper wipes, cotton swab). Carefully dry the contact with a clean, dry tissue.

Optical requirements:

Coupling loss (attenuation): 2 dB maximum per mated pair.

Weight: 8 grams maximum.

Polarization: Not applicable.

Safety wire holes: Required (three).

Force to engage and disengage:

Longitudinal force: 12 pounds maximum.

Torque: 4 inch-pounds maximum.

Coupling proof torque: 5 inch-pounds minimum.

Coupling mechanism retention force: 60 pounds minimum.

Ozone exposure: Not applicable.

Part or Identifying Number (PIN): See table I.

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TABLE I. Part or Identifying Numbers.

PIN M83522/15	Fiber size
01	200
02	400

Plug accessories: Each plug shall be supplied with suitable means for terminating and clamping a single-fiber optical cable of the fiber size and cable diameter range specified. One protective cable end cap and one set of assembly instructions shall be included. The assembly instructions shall include a cleaning procedure statement as follows:

Cleaning procedures: The lens shall be cleaned using lens tissue, a cotton swab, or a camel-hair brush and MIL-C-675 optical antireflective coating cleaning fluid.

NOTES: The plug mates with MIL-C-83522/3 adapter, MIL-C-83522/4 receptacle (active device), and MIL-C-83522/5 receptacle, bulkhead. Connection to any other type plug will result in excessive coupling loss.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Navy - SH
Air Force - 11, 17, 19, 80, 90, 99
DLA - ES

User activities:
Navy - AS, MC, OS
Air Force - 13, 14

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6060-0067)

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